CO4 - AT ①

SIMATS ENGINEERING
ASSESSMENT TOOLS

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA1301**

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 10%

QUESTIONS: 10*2=20 **Total Marks:20**

Data Interpretation

Code of Conduct:

I Om Sai Reddy T | 192372252 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Given Data (Grammar)
Consider the CFG:
S → aSb | ab

Q1. Language Identification
Based on the given grammar, which of the following strings belong to the language?
A) ab
B) aabb
C) aaabbb
D) abb

[A, B, C]

Q2. Pattern Recognition
From the grammar, identify the correct pattern:
A) $a^n b^n$
B) $(ab)^n$
C) $a^n b^m$
D) $(a+b)^*$

[A]

Q3. Derivation Analysis
Which of the following is a valid derivation for aabb?
A) S → aSb → aaSbb → aabb
B) S → ab → aabb
C) S → aSb → ab
D) S → aaSbb → aabb

[A]

Q4. Missing Step Identification
Fill in the missing step:
S → aSb → aaSbb → ______ → aabb
A) ab
B) S
C) ε
D) aSb

[A]

Q5. Parse Tree Interpretation
How many levels are there in the parse tree for string aaabbb?
A) 2
B) 3
C) 4
D) 5

[D]

Answer: D

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Consider:
S → SS | a

Q6. Ambiguity Check
This grammar is:
A) Unambiguous
~~B)~~ Ambiguous
C) Regular
D) Deterministic

[B]

Q7. String Generation
Which string can be generated?
A) a
B) aa
C) aaa
~~D)~~ All of the above

[D]

Q8. Interpretation of Ambiguity
Ambiguity means:
A) One string → one parse tree
~~B)~~ One string → multiple parse trees
C) No derivation
D) Infinite symbols

[B]

Given Data (Grammar 1)
S → 0S1 | 01

Q9. Derivation Interpretation
Which of the following derivations correctly generates the string 0011?
~~A)~~ S → 0S1 → 00S11 → 0011
B) S → 01 → 0011
C) S → 0S1 → 01
D) S → 00S11 → 0011

[A]

Given Data (Grammar 2)
S → aS | Sb | ab

Q10. Structural Analysis
Which of the following strings can be generated by the grammar?

A) ab
B) aab
C) abb
D) All of the above

[D]

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RICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understandin g	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understandin g
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandabl e with minor issues	Somewhat unclear or poorly structured	Difficult to understand

84/100